

EXAM 7 – SPRING 2012

20. (3 points)

a. (1.5 points)

Describe three reasons for an insurer to purchase reinsurance.

b. (1.5 points)

For each reason given in part a. above, explain whether proportional or non-proportional reinsurance would best satisfy that reason for purchasing reinsurance.

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Question 20 Sample Answer

Solution 1

a)

- i. Protect against large losses. May be able to hold less capital and decrease risk of insolvency, with adequate reinsurance problem from extreme events.
- ii. More stable earnings over time. Investors tend to prefer this. Also, taxes may be lower with more stable earnings.
- iii. Underwriting experience from reinsurer. Reinsurer may have more experience in a particular line and may help primary insurer be more profitable.

b)

- i. Non-proportional(excess), to protect against extreme tail events. (Proportional would decrease losses across the board, but not extreme losses by enough).
- ii. Non-proportional, since it would reduce the very large impact extreme events could have on earnings.
- iii. Proportional would probably be best here. Gives the reinsurer a stake in profitability of the line and encourages them to give good recommendations.

Solution 2

a)

- 1) Reduce catastrophe exposure - limited on largest losses
- 2) Exit line of business or geographic region
- 3) Surplus relief – need to adjust surplus by reinsurance commission

b)

- 1) Non-proportional treaty to reduce catastrophe – excess of loss
- 2) Proportional treaty that takes 100% of certain line of business or region
- 3) Proportional – Quota share to move premium and losses on % for surplus relief since also get commission premium back.

Solution 3

a)

- i. Risk mitigation: Insurer may want to lower its exposure of insolvency
- ii. Spreading risk over time: It may be more profitable for the insurer to have constant profits instead of fluctuating ones because of taxes and regulation
- iii. Achieving company goals: The company may have objections of expanding market but cannot support the rapid growth

b)

- i. Risk mitigation: non-proportional is better as it caps losses above a certain threshold, protecting the insurer from extreme losses.
- ii. Spreading risk: Non-proportional is better as high losses are capped in exchange for a lesser profit due to premium ceding
- iii. Goals: Proportional is better because it allows the insurer to recognize the reinsurance commission right away, thus decreasing its premium to capital ratio

Solution 4

a)

- i. So that they have sufficient capital to satisfy the policyholders. People want to buy insurance from a company who will be able to pay all claims made against them!
- ii. Taxation – reinsurance smoothes the income of the insurer which can assist in planning and optimizing the taxes they must pay.
- iii. Avoid cost of financial distress. By transferring risk they can avoid financial distress (where they must desperately seek to raise capital by issuing bonds with a very high yield). People also don't like to buy insurance from a company in distress, which makes the problem even worse!

b)

- i. Non-proportional to cover the extreme and unpredictable losses that could cause the insurer to go insolvent w/o reinsurance
- ii. Non-proportional to smooth out the spikes in profit and losses, thus leading to a more predictable income.
- iii. Non-proportional to cover extreme and unpredictable losses that could cause the insurer to go insolvent w/o reinsurance.

Examiner Comment

Credit for part b. was given when a solid foundational response was given for either proportional or non-proportional. In general, the non-proportional would be used to limit extreme results or the tails of distributions which may aid in stakeholder outlook, tax implications, or ratings. In general, proportional would be preferred to divest a segment of business, gain surplus relief via commissions received, and gaining insight from the reinsurer on the underlying lines of business.

For example:

To provide protection to the insurer against the effect of large losses, non-proportional reinsurance could be purchased to cover high-severity, low frequency losses which may not be fully accounted for in the underlying policy pricing.

For an unfamiliar line of business the insurer is thinking of expanding into, the purchase of proportional reinsurance coverage may provide an opportunity for the reinsurer to bring its expertise to the pricing of the policy and the managing of the claims, since it would be sharing in the fortunes (profit or loss) of the insurer.

A wide variety of possible responses was considered and as such, candidates fared particularly well on the question. For a number of the part b. responses, either proportional or non-proportional may satisfy the reason for purchase in part a., provided an appropriate explanation accompanied the selected type of insurance.

Credit was typically not given when:

1. Proportional was suggested to reduce the overall risk - this statement would be ignoring the top-line loss of premium associated with the ceded premium
2. Proportional was suggested as a means to stabilize earnings - as proportional reinsurance does not change one of the key metrics in earnings....the loss ratio
3. Non-proportional was suggested as a means to cede away the non-profitable excess layers - the excess layers may not necessarily be the unfavorable experience layers, as it depends on the associated premiums charged. Similarly, the retained layer may not be priced appropriately for the actual loss experience